

E-COMMERCE

XAML UI Layout Documentation

Visual Screens & XML Structuring Reference

Development Framework:	.NET MAUI (XAML / C#)
Architectural Pattern:	MVVM (Model-View-ViewModel)
Backend Integration:	ASP.NET Core REST APIs
Payment Gateway:	Stripe Integration with Webhooks

AppShell Navigation Structure

Description: Defines the primary app container, containing a bottom TabBar navigation schema. It sets the background colors, selected/unselected icon colors, and route points for core tabs.

Key UI Bindings:

Tab 1 -> HomePage route (default page)

Tab 2 -> CartPage route

```
<TabBar>
  <Tab Title="Home" Icon="dotnet_bot.png">
    <ShellContent
      Title="Home"
      ContentTemplate="{DataTemplate views:HomePage}"
      Route="HomePage" />
  </Tab>
  <Tab Title="Cart" Icon="dotnet_bot.png">
    <ShellContent
      Title="Cart"
      ContentTemplate="{DataTemplate views:CartPage}"
      Route="CartPage" />
  </Tab>
</TabBar>
```

1. LoginPage XAML Structure

Description: Defines the interface for user login. It contains an Error Message block that handles real-time warning indicators, input Entry containers for Email/Password, and the primary Action button.

Key UI Bindings:

HasError -> Controls validation box visibility

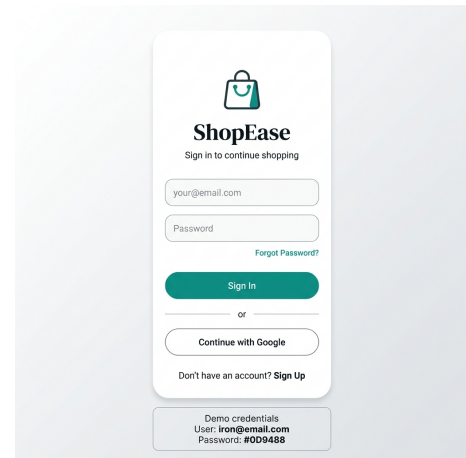
Email -> Entry field binding

Password -> Password masking input

LoginCommand -> Button trigger action

```
<!-- Login Button -->
<Button Text="Sign In"
        BackgroundColor="{StaticResource Primary}"
        TextColor="White" FontAttributes="Bold"
        FontSize="16"
        Margin="0,28,0,0"
        BorderRadius="14" HeightRequest="54"
        Command="{Binding LoginCommand}" />

<!-- Loading Indicator -->
<ActivityIndicator IsRunning="{Binding IsBusy}"
        IsVisible="{Binding IsBusy}"
        Margin="0,12,0,0" />
        Color="{StaticResource Primary}"
```



2. HomePage XAML Structure

Description: Features the main grid display. Highlights dynamic welcome labels, action icons (Cart, Settings), and a live searching framework. Displays categories as horizontal chips with active visual state triggers.

Key UI Bindings:

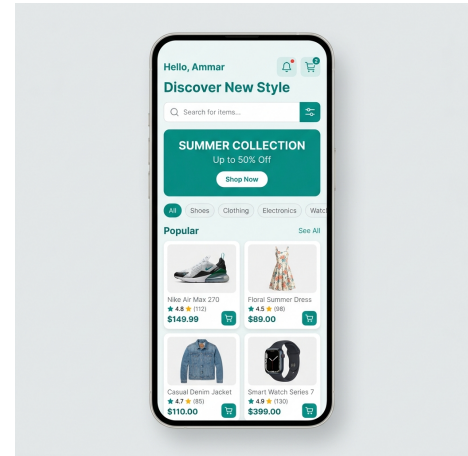
UserName -> Displays personalized header

SearchTerm -> Live search text entry

Categories -> CollectionView of scrollable category chips

ToggleCategoriesCommand -> Collapses/expands chips

```
<CollectionView ItemsSource="{Binding Categories}"
HeightRequest="44" Margin="14,0" IsVisible="{Binding
ShowCategories}">
    <CollectionView.ItemsLayout>
        <LinearItemsLayout
Orientation="Horizontal" ItemSpacing="8" />
    </CollectionView.ItemsLayout>
    <CollectionView.ItemTemplate>
        <DataTemplate
x:DataType="models:Category">
            <Frame CornerRadius="14"
Padding="14,8" HasShadow="False" BorderColor="Transparent">
                <Frame.Style>
                    <Style TargetType="Frame">
                        <Setter
Property="BackgroundColor" Value="{StaticResource CardBackground}"
/>
                    </Style.Triggers>
                    <DataTrigger
TargetType="Frame" Binding="{Binding IsSelected}" Value="True">
                        <Setter
Property="BackgroundColor" Value="{StaticResource Primary}" />
                    </DataTrigger>
                </Style.Triggers>
            </Frame>
        </DataTemplate>
    </CollectionView.ItemTemplate>
</CollectionView>
```



3. ProductDetailsPage XAML Structure

Description: Details the layout for single product items. Displays image, description, stock indicator, rating chips, and a sticky bottom grid holding the total price calculation and the primary cart action button.

Key UI Bindings:

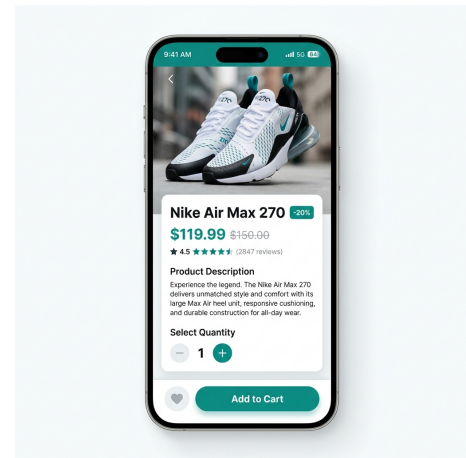
Product.Image -> Renders product image

TotalPrice -> Real-time quantity-multiplied cost

CartButtonText -> Dynamic CTA label ('Add to Cart' or 'Go to Cart')

AddToCartCommand -> Submits data to CartService

```
<!-- Add/Go to Cart CTA -->
<Button Grid.Column="1" Text="{Binding
CartButtonText}"
        BackgroundColor="{StaticResource Primary}"
        TextColor="White" FontAttributes="Bold"
        FontSize="15"
        CornerRadius="16" HeightRequest="54"
        Command="{Binding AddToCartCommand}" />
</Grid>
</Frame>
```



4. CartPage XAML Structure

Description: Shows list of items added to the cart inside a SwipeView container (which handles quick swiping deletion). It includes dynamic order totals (Subtotal, Shipping, and Final Price) at the bottom.

Key UI Bindings:

CartItems -> List source bindings

Subtotal / TotalAmount -> Cost summaries

CheckoutCommand -> Submits orders to checkout flow

RemoveItemCommand -> Triggers swiped item deletion

```
<SwipeView Margin="0,0,0,12">
  <SwipeView.RightItems>
    <SwipeItems Mode="Execute">
      <SwipeItem Text="Delete"
        BackgroundColor="{StaticResource Danger}"
        Command="{Binding
          Source={RelativeSource AncestorType={x:Type vm:CartViewModel}},
          Path=RemoveItemCommand}"
        CommandParameter="{Bind
ing .}" />
    </SwipeItems>
  </SwipeView.RightItems>
```

